I am a labor economist working in education, public, and behavioral economics. Three primary themes connect my research. The first is a focus on decision making under uncertainty, in particular allowing for subjective beliefs which may reflect biases and information frictions. This involves designing surveys and collecting novel data. The second is on public policy and labor market outcomes. The third is on applied econometrics: across these projects, I develop and use new methods to measure beliefs, analyze administrative data, and integrate both into economic models.

My job market paper, *Beliefs and Preferences in Community College*, studies the role that subjective beliefs about earnings, tuition costs, and dropout risks play in the high dropout and low four-year transfer rates of community college students. In order to address this, I collected novel survey data from a large Maryland institution on students' beliefs and expected paths through higher education. I find evidence against full-information rational expectations; students are pessimistic about tuition costs, and optimistic about dropout risk, with fairly well-calibrated beliefs about earnings. I use an information experiment to study updating behavior and identify the causal effect of beliefs on students' higher education plans. I then use the expectations data to estimate a discrete choice model, which allows me to study the importance of each belief and the impact of correcting biases. Students are most responsive to four-year dropout risk, and inelastic to expected tuition costs, with substantial welfare benefits from correcting beliefs. I show that policies aimed at lowering the four-year dropout risk (for example, through intensive counseling services for transfer students) are likely to be more effective than tuition subsidies. Finally, my model suggests that students' learning about their match with school is likely an important factor in dropout and transfer.

Where possible, I introduce methodological advancements for better use of expectations data. In my job market paper, I develop a maximum likelihood estimator which allows for measurement error in the dependent variable. In the third chapter of my dissertation (second chapter described below), *Identifying Measurement Error in Subjective Beliefs*, I propose a method to allow for measurement error in the full set of elicited beliefs, an issue that much of this literature sidesteps. I use the control group from the information experiment to model the error: pre- and post-elicitations allow me to control for classical error, while clusters of similar types of beliefs allow for identification of non-classical error. I then show that the treatment effects estimator and estimates of learning over time are sensitive to this error.

I am continuing my interests in non-standard beliefs and data collection in early work with coauthors from Johns Hopkins and the American Enterprise Institute. We have partnered with the International Organization for Migration to field a large-scale survey on the subjective expectations of prospective migrants in El Salvador (planned for late 2026). The survey instrument has been designed with input from researchers across disciplines at both JHU and the AEI, and the data will be a shared resource for researchers interested in migration policy at both institutions. The final product will be a unique longitudinal dataset that will enable us to study how intentions map onto outcomes and how information spreads geographically, and to estimate a structural model of return migration.

Another chapter of my dissertation (coauthored with JHU alumnus Wonsik Ko and JHU Ph.D. candidate Su Wang) is *Labor Market Outcomes from Workforce Training Programs*, and reflects my interests in education, empirical methods, and public economics. We use Maryland administrative data to study how workforce training affected the unemployment scarring effect for young workers

who lost their jobs in the COVID-19 recession. Because our administrative data only contains information on total earnings, we develop a method which uses machine learning tools and auxiliary data to impute wages and hours for our sample. This allows us to examine the impact of training on human capital (measured by the wage) separately from its impacts on labor supply, providing a more nuanced analysis than prior papers in this literature typically provide.

Two projects with my advisor, Dr. Robert Moffitt, further showcase my work in public economics and interest in inequality. With JHU alumnus Emma Kalish, we examine changes in the tax-and-transfer system from 1983-2019. We calculate benefits and marginal tax rate schedules for a comprehensive set of programs, demonstrating what we term the "Great MTR Pivot": marginal rates have fallen at low levels of earnings, but increased at higher levels. We estimate a static model of labor supply and program participation which takes account of this full set of programs. We allow for incomplete takeup due to stigma, paperwork costs, and other non-pecuniary factors, and calibrate the model to match the best causal estimates from the literature of each program's impact. We show that these reforms indeed increased labor supply and welfare, but caused increased inequality. We also calculate alternative (quasi-)optimal schedules.

Further pursuing the study of inequality, in early work with Daniel Rodriguez, Dr. Moffitt and I use IRS tax records to look at changes in the bottom quartile of the earnings distribution. Although many papers have examined earnings inequality by focusing on the top earners, relatively little work has looked at the lowest earners. The few such studies that do this have generally used the CPS, which suffers from a small sample and measurement error that make examining fine-grained percentile changes difficult. We use IRS records and correct for non-filing and under-reporting of income to obtain a more complete picture of how inequality has evolved.

Going forward, I am eager to continue my work in public economics and the economics of education. For example, Brian Murphy (Ph.D. student at the University of Houston) and I have formulated a proposal to use Texas administrative data to study the way that post-closure teacher reallocation can exacerbate education quality disparities both within and across districts. I am also excited to build my behavioral economics agenda through continued collection and use of subjective expectations data. Data collection at the community college is ongoing; I also have plans to link the survey data to state administrative data, which will allow for a richer analysis of realized outcomes and for a study of how early childhood experiences can influence beliefs later in life. I would also like to begin new survey projects. With Jamie Hyder (Ph.D. student at JHU), I am developing a method for eliciting expectations in circumstances of deep uncertainty which we plan to test with an application to how college students evaluate their career prospects in the face of advances in artificial intelligence. Across all my work, I plan to continue combining novel survey data and administrative data to study how individual beliefs and public policy shape education and labor market outcomes.